

S.No. : 255

NEE 4101

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Following Paper ID and Roll No. to be filled in your Answer Book.

PAPER ID : 43301

Roll No.

No.

B. Tech. Examination, 2025-26

(Odd Semester)

BASIC ELECTRICAL ENGINEERING

Time : Three Hours]

[Maximum Marks : 60

Note :- Attempt all questions.

SECTION - A

1. Attempt all parts of the following : $8 \times 1 = 8$

- (a) Define bandwidth.
- (b) Define bilateral and unilateral elements with example.
- (c) Define form factor.
- (d) Calculate r.m.s. value for $V = 416 \sin \omega t$.

[P. T. O.]

- (e) Write two application of auto-transformer.
- (f) Define hysteresis loss.
- (g) Synchronous motor is self starting or not?
- (h) Draw impedance triangle for RLC series ckt.

SECTION – B

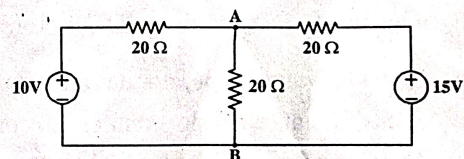
2. Attempt any two parts of the following : $2 \times 6 = 12$

- (a) Develop the relationship for delta to star and star to delta transformation.
- (b) Describe PMMC type instruments in detail. Also discuss its advantages and disadvantages.
- (c) Find average and r.m.s. value of pure sinusoidal wave.
- (d) Explain why single phase induction motor is not self starting. Explain various method to start it.

SECTION – C

Note :- Attempt all questions. Attempt any two parts from each questions. $8 \times 5 = 40$

- 3. (a) State and explain Norton's theorem to solve network problems. Mention two advantages of Norton's theorem.
- (b) State and prove maximum power transfer theorem.
- (c) State and explain Thevenin's theorem. Calculate the current in branch AB in given circuit using Thevenin's theorem :



- 4. (a) What is resonance in R-L-C series circuit? Show that the resonant frequency :

$$f_r = \sqrt{f_1 f_2}$$

- (b) Prove that the average power consumed in pure inductive circuit is zero.
- (c) A choke coil having a resistance of 15Ω and inductance of 0.05 H is connected in series with a condenser of $150 \mu\text{F}$. The whole circuit has been connected to 250 V , 50 Hz . Calculate impedance, current and power factor.

[P.T.O.]

5. (a) State Flemings left hand, Fleming right hand and Lenz law.
- (b) Explain the working of induction type single phase energy meter with neat diagram.
- (c) Derive e.m.f. equation of a single phase transformer.
6. (a) Give constructional detail of DC motor.
- (b) Derive torque equation of DC motor.
- (c) A wave wound generator having four pole and it has 51 slots each slot contains 20 conductors. What will be the voltage generated in the machine when driven at 1500 r.p.m. assuming the flux per pole to be 7.0 mwb?
